

Amazon "Add to Cart" & "Checkout" Features					
Test Cases:					
Test Case ID	Test Scenario	Test Steps	Expected Result	Status	
tc01	Verify adding an in-stock item to the Cart	1. Open a product page. 2. Click the "Add to Cart" button	The item is added successfully, and the Cart count increases by +1	PASS	
tc02	Verify adding an out-of-stock item	1. Open an out-of-stock product page. 2. Check the "Add to Cart" button.	The "Add to Cart" button should be disabled or display "Out of Stock"	PASS	
tc03	Verify updating item quantity in the Cart	1. Open the Cart page. 2. Change item quantity from 1 to 3.	Total price updates automatically to reflect the quantity of 3	PASS	
tc04	Verify removing an item from the Cart	1. Open the Cart page. 2. Click the "Delete/Remove" button next to the item.	The item is removed from the Cart, and the total price decreases	PASS	

tc05	Verify Cart items after logging out and logging back in	1. Add an item to the Cart. 2. Log out and log back in	The previously added item should still remain in the Cart.	PASS		
tc06	Verify Checkout process with a valid saved address	1. Click "Proceed to Checkout". 2. Select an existing delivery address.	System redirects to the payment page successfully.	PASS		
tc07	Verify adding a new delivery address during Checkout	1. On the Checkout page, select "Add New Address". 2. Enter valid address details and click Save	New address is saved and selected for delivery	PASS		
tc08	Verify payment using a valid Credit Card	1. Select Credit Card payment option. 2. Enter valid card details, OTP, and confirm.	Payment completes successfully and Order Confirmation is shown.	PASS		

tc09	Verify payment using an expired Credit Card	1. Select Credit Card payment option. 2. Enter details of an expired card.	System displays an error message "Invalid/Expired Card" and payment fails	FAIL		
tc10	Verify Order Summary calculations before placing an order	1. Navigate to the final Checkout page. 2. Check product price, delivery fee, and taxes	Total amount calculation (Items + Delivery + Taxes) must be accurate	PASS		
Testing Metrics:						
Problem Statement:						
Total Project Requirements: 10						
Requirements Tested: 9						
Total Test Cases Created: 10						
Test Cases Executed: 10						
Passed Test Cases: 9						
Failed Test Cases: 1						
Lines of Code (LOC): 5,000 lines						
Total Defects Found: 1 defect						
Critical Defects Found: 0						
High Severity Defects Found: 1 defect (Expired Credit Card Payment Validation)						
Checkout Page Load Times (5 Runs): 2 sec, 3 sec, 2 sec, 4 sec, 19 sec						

Answers:						
Test / Requirement Coverage % :						
		$(\text{Requirements Tested} \div \text{Total Requirements}) \times 100$				
		Test coverage= $(9 \div 10) \times 100 = 90\%$				
Test Execution Rate %:						
		$(\text{Executed Test Cases} \div \text{Total Test Cases}) \times 100$				
		rate= $(10 \div 10) \times 100 = 100\%$				
Pass Rate % :						
		$(\text{Passed Test Cases} \div \text{Executed Test Cases}) \times 100$				
		rate= $(9 \div 10) \times 100 = 90\%$				
Fail Rate%:						
		$(\text{Failed Test Cases} \div \text{Executed Test Cases}) \times 100$				
		rate= $(1 \div 10) \times 100 = 10\%$				
Defect Density (defects per KLOC) % :						
		Number of Defects $\div$ KLOC				
		rate= $1 \div 5 = 0.2$				
High Severity Defect Percentage % :						
		$(\text{High Severity Defect} \div \text{Total Defect}) \times 100$				

	rate=(1÷1)*100=100%				
Performance Metrics:					
	Sum of Values ÷ Number of Values				
	Mean: (2+3+2+4+19)÷5= 6 sec				
	Median:3 sec				